



# Applied Machine Learning

## Benchmarking: Large-Scale Benchmarking & OpenML



OpenML

### Learning goals

- Why Benchmarking?
- What is OpenML?

# WHY BENCHMARKING



It is hard to evaluate methods using theoretical/mathematical analysis alone.

- Theoretical analysis typically requires strong assumptions (e.g., infinite sample size, no noise, ideal model class).
- Implementation details cannot be formally compared through purely theoretical/mathematical analysis (e.g., runtime, memory use).
- Effect of different data properties on algorithms cannot be explained theoretically, e.g., does XGBoost outperform an LM on data with only categorical features?
- Hyperparameter settings are often not part of the theoretical analysis.

# THE STEPS OF BENCHMARKING



- 1 Formulate a hypothesis (about algorithm performance or behavior)
- 2 Define an appropriate performance metric (if not part of the hypothesis)
- 3 Select or collect datasets to evaluate the hypothesis
- 4 Define a resampling strategy
- 5 Define hyperparameter search space and tuning strategy
- 6 Execute the full cross-product of learners, datasets, and tuning configurations
- 7 Analyze results and draw conclusions

# THE PROBLEM WITH DATASETS

- No universally adopted standard dataset format
- No universally adopted dataset sharing platform
- Few platforms offer programmatic (API-based) access to data



# THE IDEAL WORLD - FRICTIONLESS ML



## Datasets

- Unified access to all datasets via unified API
- Possibility to share own data (mostly relevant in academic context)
- Discover and search datasets via their meta-data

## Associated Objects

- **Tasks:** Predefined train/test splits for standardized evaluation
- **Runs:** Publicly shared results (algorithm, settings, performance)

# OPENML - THE PROJECT

► *OpenML.org* n.d.

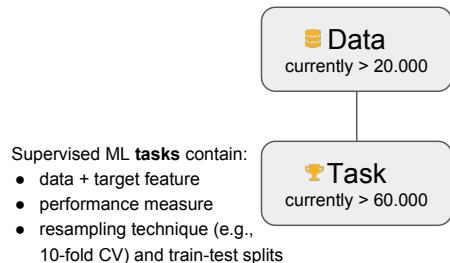


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**Benchmarking:** compare algorithms w.r.t. performance/runtime on datasets.

**Goal:** Reproducibility, transparency, and large-scale collaboration in ML.

OpenML relies on 4 **basic components** (just like benchmark experiments):



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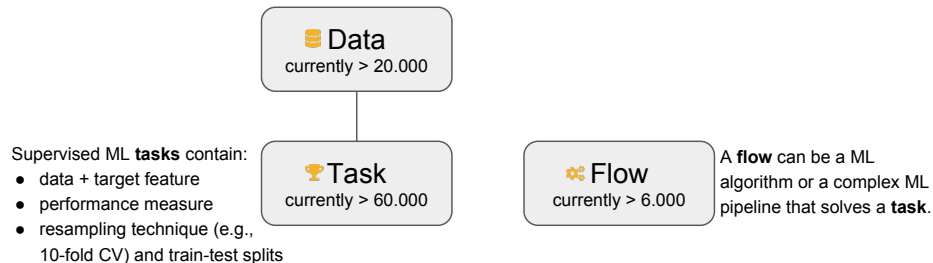


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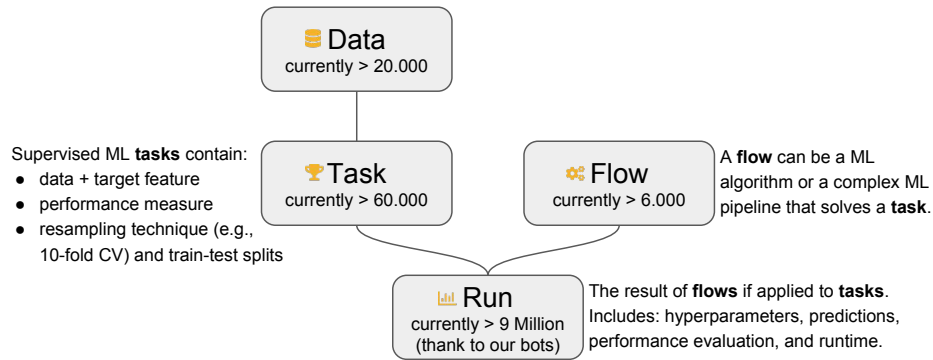


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**Note:** *OpenML also supports **collections** of tasks and runs*





- **Raw data** stored in standardized formats:
  - .arff (Attribute-Relation File Format): plain-text format with inline schema
  - .parquet: efficient binary columnar storage (preferred for large data)
- **Meta-data** annotations include:
  - Dataset description (source, intended use, licensing)
  - Feature names and types (numeric, categorical, ordinal, date, string)
  - Designated roles: input features, target variable, ignored columns
- **Automatically extracted information:**
  - Meta-features (e.g., number of instances, number of features)
  - Statistical summaries (missing values, sparsity, imbalance)
  - Baseline performance of simple models (e.g., majority class classifier)

# OPENML TASKS

- A **task** defines a concrete ML problem on a dataset, e.g., in supervised ML:
  - Target feature (to be predicted)
  - Resampling strategy (e.g., 10-fold CV, holdout)
  - Evaluation metric (e.g., accuracy, RMSE)
- Tasks ensure reproducibility by fixing all relevant components
- **Flows** (i.e., ML pipelines or learners) are executed on *tasks* and produce *runs*



# OPEN QUESTION: WHAT DATASETS/TASKS TO USE?



- Hard to compare results across papers
- Benchmarking is often done on small set of datasets
  - Question about generalization to other datasets
  - Cherry picking or arbitrary selection
  - Different versions, different train-test setups
- Publication bias
  - Published papers report good results
  - Interesting to know WHEN an algorithm works (and when it doesn't)

# OPENML BENCHMARKING SUITES



- **Benchmarking suite:** curated collection of standardized OpenML tasks
- Provide unified, shareable task definitions with fixed resampling and target
- Accessible via APIs in R, Python, and Java
- Enable reproducible and comparable experiments toward shared research goals

# EXAMPLE I: OPENML-CC18 (CLASSIFICATION)

► “Bischl et al.” 2021



Standardized benchmark suite for evaluating supervised classification algorithms.

- 72 classification tasks
- Medium size: 500–100,000 observations,  $\leq 5000$  features (post encoding)
- Contains missing values and categorical features
- Excludes:
  - Strong class imbalance
  - Time series or group structure
  - Sparse data and free-form text
- Follows objective and subjective selection criteria (see paper)

# EXAMPLE II: OPENML-CTR23 (REGRESSION)

► “Fischer et al.” 2023



Extension of CC18 selection principles to regression tasks.

- Uses same structural filters as CC18
- Includes only datasets with numeric targets and  $\geq 5$  distinct values
- Excludes:
  - Datasets trivially solved by linear models
  - Datasets with ethical/legal concerns
  - Datasets restricted from public benchmarking

# EXAMPLE III: AUTOML BENCHMARK SUITE (AMLB)

► “Gijssbers et al.” 2024



Designed to evaluate modern AutoML systems on realistic and diverse problems.

- 71 classification tasks and 33 regression tasks
- Stricter inclusion criteria for difficulty and real-world complexity
- Covers multiple domains; avoids trivial, overly cleaned, or synthetic data
- Excludes:
  - Free-text features
  - Time dependencies or metadata leaks

# WHY PARALLELIZATION MATTERS



## Benchmark Scenario (AutoML Study)

- 71 datasets  $\times$  10-fold cross-validation
- 9 AutoML systems evaluated
- 1 hour per system-dataset-fold combination

⇒ Total runtime: **6390 hours** = **266+ days** sequentially

**Conclusion:** *Benchmarking at scale is infeasible without parallelization.*